

ADMM — Assumption Dependency Mapping Module

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

Parent Standard:

Prediction Assumption Standard (PAS)

Operational Layer: Prediction Assumption Governance Layer

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Type: Parent Standard Module

Governed Space:

Assumption Dependency Mapping

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Compatibility: OOF® Methodology OS™ · GOA™ · ORA™ · VALIDOS™ · INTEGROS® · ADIT® · AGA™ · AIG® · ASGA™ · CLIA® · SIMULOS®

AI-Readable: Yes

Authority: OOF®

Protection:

MIP® — Methodological Intellectual Property

Canonical Language:

English (UCL™)

Governed Space

Assumption Dependency Mapping

Canonical Definition

Assumption Dependency Mapping Module (ADMM) defines the governance conditions under which identified Predictive Assumptions are mapped to the assumptions, Prediction Inputs, methodology requirements, operating conditions, system states, external conditions, decisions, and other predictive dependencies upon which they rely or which rely upon them.

ADMM establishes the governed dependency structure of Prediction Assumptions.



transforms a list of assumptions into a relational structure showing:

what depends on what

and:

where assumption change may propagate through the predictive system.

ADMM does not determine how sensitive the prediction is to an assumption, how material the assumption is, or what governance action must follow when it changes.

Those functions belong to later PAS modules.

Its function is to make dependency visible.

Assumption Identification $\neq$ Assumption Dependency Governance

Operational Role

ADMM serves as the second internal governance space of the Prediction Assumption Standard (PAS) .

It receives:

Identified Predictive Assumptions

from:

PAIM — Predictive Assumption Identification Module

and establishes their material dependency relationships.

The module asks:

What does each assumption depend upon, what depends upon it, and how could a change in one assumption affect other parts of the predictive structure?

The governed progression is:

Identified Predictive Assumptions → Dependency Identification → Dependency Classification → Assumption Chain Mapping → Dependency Concentration → Dependency State → Governed Assumption Dependency Map

The resulting dependency map becomes an input to:

ASNM — Assumption Sensitivity Module.

Internal Module Flow

Predictive Assumption Identification → → Assumption Dependency Mapping → Assumption Sensitivity → Assumption Materiality → Assumption Change Governance

Current Position: Assumption Dependency Mapping

Module Operational Space

ADMM governs:

assumption-to-assumption dependency; assumption-to-input dependency; assumption-to-methodology dependency; assumption-to-operational-condition dependency; assumption-to-system-state dependency; assumption-to-external-condition dependency; assumption-to-decision dependency; assumption-to-resource dependency; direct dependency; indirect dependency; upstream dependency; downstream dependency; conditional dependency; shared dependency; common dependency; dependency chain; dependency cluster; dependency network; dependency concentration; dependency depth;

dependency direction; dependency state; dependency uncertainty; dependency conflict; circular dependency; dependency bottlenecks; dependency propagation paths; dependency change; and historical dependency preservation.

Module Function

A prediction may contain many assumptions that appear independent while actually relying on the same underlying condition.

For example:

Supplier remains operational

may support:

Components remain available

which supports:

Production remains uninterrupted

which supports:

Delivery targets remain achievable.

If the supplier fails, several downstream assumptions may fail together.

Without dependency mapping, the prediction may appear to rely on several independent assumptions when in reality all of them depend upon one upstream condition.

ADMM makes that structure visible.

Multiple Assumptions $\neq$ Independent Assumptions

Assumption Dependency

An Assumption Dependency exists where the applicability, plausibility, meaning, state, or continued relevance of one assumption materially depends upon another governed element.

That element may be:

another assumption; Prediction Input; methodology requirement; operational condition; system state; external condition; decision; resource; boundary; time condition.

Direct Dependency

A Direct Dependency exists where one assumption materially relies upon another element without a materially relevant intermediary.

Example:

Assumption A: Supplier remains operational

↓

Assumption B: Supplier continues deliveries

Assumption B directly depends upon Assumption A.

Indirect Dependency

An Indirect Dependency exists where a relationship operates through one or more intermediary assumptions or conditions.

Example:

Supplier Solvency

↓

Supplier Operations

↓

Component Availability

↓

Production Continuity

The relationship between supplier solvency and production continuity may be indirect but highly important.

Indirect Dependency $\neq$ Weak Dependency by Default

Upstream Dependency

An Upstream Dependency is a condition or assumption upon which another governed assumption depends.

Upstream dependencies may become critical because failure can propagate across multiple downstream assumptions.

Downstream Dependency

A Downstream Dependency is an assumption, predictive condition, or predictive state affected by an upstream dependency.

Mapping downstream relationships helps determine where an assumption change could propagate later in the Prediction Lifecycle.

Dependency Direction

Dependency should remain directionally identifiable where material.

For example:

$A \rightarrow B$

does not automatically imply:

$B \rightarrow A$

Dependency $A \rightarrow B \neq$ Dependency $B \rightarrow A$

This distinction prevents false assumptions of mutual dependence.

Conditional Dependency

Some dependencies exist only under defined conditions.

Example:

Demand Forecast Depends on Weather

only where:

Outdoor Event Condition = Active

Conditional dependency must not be represented as universal dependency.

Conditional Dependency $\neq$ Universal Dependency

Shared Dependency

Two or more assumptions may depend upon the same underlying condition.

Example:

Assumption A

and:

Assumption B

both depend upon:

System X remains operational.

This Shared Dependency can create concentrated predictive vulnerability.

Common Dependency

Common Dependency represents a materially shared underlying requirement across multiple assumptions.

It is especially important where apparently separate assumptions are only conditionally independent.

Separate Assumption Statements $\neq$ Separate Dependency Foundations

Assumption Chain

An Assumption Chain is a sequence of materially linked assumptions or conditions.

Example:

Regulation Remains Stable

↓

Product Remains Legally Marketable

↓

Distribution Continues

↓

Projected Sales Remain Achievable

ADMM preserves the chain so that later sensitivity and materiality assessment can examine the consequences of change.

Assumption Dependency Network

Complex predictions may involve many interconnected assumptions.

The resulting structure may be represented as an Assumption Dependency Network .

A network may contain:

upstream dependencies; downstream dependencies; shared dependencies; branching dependencies; converging dependencies; conditional dependencies; circular relationships.

ADMM does not require excessive modeling complexity.

It requires enough structure to identify material dependency relationships.

Dependency Cluster

A Dependency Cluster is a group of assumptions materially linked through one or more shared dependencies.

Clusters may be useful where predictive vulnerability is concentrated around:

one supplier; one market condition; one infrastructure system; one policy state; one external actor; one methodology condition.

Dependency Concentration

Dependency Concentration occurs where many assumptions depend upon a limited number of upstream conditions.

High dependency concentration can make a prediction structurally fragile.

It does not itself establish materiality.

That is governed later by AMAM.

Dependency Concentration $\neq$ Assumption Materiality by Default

Dependency Depth

Dependency Depth represents the number or complexity of dependency layers between an assumption and its underlying conditions.

Deep dependency chains may make predictive structure harder to understand and monitor.

Depth alone does not establish risk.

Dependency Bottleneck

A Dependency Bottleneck exists where one upstream condition supports many downstream assumptions.

For example:

Primary Data Feed Remains Available

may support:

market-state assumption; price-stability assumption; liquidity assumption; demand assumption; execution-condition assumption.

A bottleneck may become a critical structural point in later sensitivity and materiality analysis.

Assumption-to-Input Dependency

An assumption may materially depend upon Prediction Basis.

Example:

Demand remains stable

may be supported by recent transaction data.

ADMM maps the relationship:

Prediction Input $\rightarrow$ Assumption

without converting the assumption into fact.

Input Support $\neq$ Assumption Truth

Assumption-to-Methodology Dependency

A methodology may require certain assumptions to hold.

Example:

historical relationship remains applicable

may be necessary for a forecasting methodology to remain meaningful.

ADMM maps this relationship.

PMES retains ownership of Prediction Methodology.

Assumption-to-Operational-Condition Dependency

An assumption may depend upon an operational condition.

Example:

Delivery capacity remains sufficient

may depend upon:

Warehouse automation remains operational.

ADMM maps the predictive dependency.

ORA™ governs Operational Reality itself.

Assumption-to-System-State Dependency

AI, software, industrial, and autonomous systems may contain assumptions dependent upon system states such as:

model version; operating mode; system configuration; network availability; permission state; infrastructure state.

A change in system state may therefore affect multiple assumptions.

Assumption-to-External-Condition Dependency

External dependencies may include:

regulation; interest rates; weather; geopolitical conditions; supplier state; infrastructure; commodity availability.

Externality does not reduce governance importance.

Assumption-to-Decision Dependency

A predictive assumption may depend upon a decision made by:

management; regulator; operator; customer; supplier; government; algorithmic system; autonomous agent.

The decision may not yet be known.

The assumption must still preserve the dependency where material.

Assumption-to-Resource Dependency

Some assumptions rely upon continuing availability of:

capital; personnel; energy; materials; computing capacity; logistics; infrastructure.

These dependencies can become critical in operational predictions.

Dependency Independence

An assumption should not be treated as independent merely because no explicit dependency has yet been documented.

No Documented Dependency ≠ Demonstrated Independence

Where independence materially affects predictive interpretation, it should be sufficiently supported.

Dependency Uncertainty

Dependency relationships may themselves be uncertain.

Potential states may include:

Established

Probable

Possible

Uncertain

Not Materially Established

ADMM preserves uncertainty rather than forcing a deterministic dependency map.

Dependency Strength

Where methodology supports it, dependency may be represented as:

Weak

Moderate

Strong

Critical

or other governed states.

Dependency strength should not be confused with assumption materiality.

Strong Dependency $\neq$ High Materiality by Default

Circular Dependency

Circular Dependency exists where assumptions materially depend upon one another in a loop.

Example:

Market Confidence Remains Strong

supports:

Investment Continues

which supports:

Market Confidence Remains Strong.

Circular structures require explicit visibility because they can create misleading predictive reinforcement.

Mutual Support $\neq$ Independent Confirmation

Feedback Dependency

Some predictive environments contain dynamic feedback relationships.

A prediction may depend upon assumptions that are themselves affected by predictive outputs, decisions, or actions.

Where material, ADMM identifies the dependency relationship.

Prediction-level reflexivity remains governed through the broader PGA™ lifecycle.

Dependency Conflict

A Dependency Conflict may occur where assumptions depend upon conditions that cannot simultaneously hold or where one dependency undermines another.

For example:

Assumption A requires low prices

while:

Assumption B requires high supplier margins.

Such structural tension must remain identifiable.

Dependency Change

Dependencies may change over time.

Examples include:

supplier replacement; methodology change; system redesign; new regulation; organizational restructuring; change in decision authority; new information pathway.

A historical dependency map must not be silently overwritten.

Dependency State

A governed dependency may possess a state such as:

Active

Conditional

Uncertain

Weakened

Broken

No Longer Applicable

The applicable state depends upon implementation context.

Dependency Breach

A dependency may cease to support the assumption as expected.

This does not itself determine that the downstream assumption has failed.

It creates a governance condition requiring later sensitivity and change analysis.

Dependency Breach $\neq$ Automatic Assumption Failure

Dependency Propagation Path

A Dependency Propagation Path identifies how a change in one assumption or condition may affect other assumptions.

Example:

Fuel Supply Disruption

↓

Transport Capacity Assumption Weakens

↓

Delivery Continuity Assumption Weakens

↓

Revenue Forecast Assumption Weakens

ADMM maps the pathway.

ASNM later assesses sensitivity.

ACGM later governs actual change propagation.

Dependency Isolation

An assumption may be relatively isolated from other assumptions.

Isolation should not be assumed without examining material dependencies.

An isolated assumption may still be highly material.

Historical Dependency State

The dependency structure applicable at prediction formation must remain distinguishable from later dependency relationships.

For example, a supplier may later become vertically integrated into the organization.

That later relationship must not rewrite the historical dependency state.

Current Dependency Structure $\neq$ Historical Dependency Structure

Dependency Mapping Sufficiency

Assumption Dependency Mapping is sufficiently governed when:

material dependency relationships are identifiable; dependency direction is clear; direct and indirect dependencies are distinguishable where necessary; shared dependencies are visible; dependency chains can be reconstructed; dependency concentration is identifiable; circular dependencies remain visible; dependency uncertainty is preserved; material propagation paths can be traced; dependency changes remain historically reconstructable.

Governance Requirements

ADMM requires that:

1. identified Predictive Assumptions be assessed for material dependencies; 2. assumption-to-assumption dependencies be identifiable; 3. assumption-to-input dependencies remain distinguishable; 4. assumption-to-methodology dependencies be represented where material; 5. operational and system-state dependencies remain visible; 6. external-condition dependencies be identified where material; 7. dependency direction remain explicit; 8. direct and indirect dependencies remain distinguishable where necessary; 9. conditional dependencies not be represented as universal; 10. shared dependencies remain visible; 11. dependency chains be reconstructable; 12. dependency concentration be identifiable; 13. no documented dependency not be treated as demonstrated independence; 14. dependency uncertainty remain explicit; 15. circular dependencies remain visible; 16. mutual support not be mistaken for independent confirmation; 17. dependency conflicts remain governable; 18. dependency breach remain distinguishable from automatic assumption failure; 19. material propagation paths remain identifiable; 20. current dependency structure remain distinguishable from historical dependency structure.

Governance Logic

ADMM applies the following governance logic:

Identified Predictive Assumptions

↓

Potential Dependencies Identified

↓

Assumption-to-Assumption Relationships Mapped

↓

Input / Methodology / Operational / System / External Dependencies Mapped

↓

Dependency Direction Established

↓

Direct / Indirect / Conditional Relationships Classified

↓

Shared Dependencies Identified

↓

Assumption Chains and Clusters Mapped

↓

Dependency Concentration / Bottlenecks Identified

↓

Circular Relationships and Conflicts Identified

↓

Dependency State and Uncertainty Established

↓

Propagation Paths Preserved

↓

Governed Assumption Dependency Map

↓

ASNM — Assumption Sensitivity Module

Where dependency relationships remain materially unresolved:

Assumption Dependency Governance Gap → Clarification / Additional Mapping / Qualification / Explicit Uncertainty

Minimum Implementation Framework

A minimum implementation of ADMM must establish:

A. Assumption Reference

The Predictive Assumption governed through PAIM.

B. Dependency Identifier

A sufficiently unique identifier for the dependency relationship.

C. Dependency Target

The assumption, input, methodology requirement, condition, system state, decision, resource, or other governed element upon which the assumption depends.

D. Dependency Type

At minimum:

Assumption / Input / Methodology / Operational Condition / System State / External Condition / Decision / Resource / Other

E. Dependency Direction

At minimum:

Upstream → Downstream

F. Dependency Form

Where applicable:

Direct / Indirect / Conditional / Shared / Circular

G. Dependency State

At minimum:

Active / Conditional / Uncertain / Weakened / Broken / No Longer Applicable

H. Dependency Strength

Where used:

Weak / Moderate / Strong / Critical / Unable to Determine

I. Shared Dependency Reference

Where several assumptions rely upon the same underlying condition.

J. Dependency Chain Reference

Where the assumption participates in a wider dependency chain.

K. Propagation Path

The material downstream relationships potentially affected by dependency change.

L. Conflict / Circularity Flag

Any material structural conflict or circular relationship.

M. Historical Dependency State

The dependency relationship applicable at prediction formation and material later changes.

Governance Outputs

ADMM may produce:

Governed Assumption Dependency Map; Assumption Dependency Register; assumption-to-assumption map; assumption-to-input map;

methodology dependency record; operational dependency record; system-state dependency record; external-condition dependency record; decision dependency record; resource dependency record; Shared Dependency Map; Assumption Chain Map; Dependency Cluster Map; Dependency Concentration Profile; Dependency Bottleneck Record; Circular Dependency Record; Dependency Conflict Record; Dependency Propagation Map; dependency-state record; dependency-uncertainty record; historical dependency record; unresolved dependency requirements.

The primary governed output is:

Governed Assumption Dependency Map

This becomes a principal input into:

ASN — Assumption Sensitivity Module.

Critical Distinctions

Assumption Identification $\neq$ Assumption Dependency Governance

Multiple Assumptions $\neq$ Independent Assumptions

Indirect Dependency $\neq$ Weak Dependency by Default

Dependency $A \rightarrow B \neq$ Dependency $B \rightarrow A$

Conditional Dependency ≠ Universal Dependency

Separate Assumption Statements ≠ Separate Dependency Foundations

Dependency Concentration ≠ Assumption Materiality by Default

Input Support ≠ Assumption Truth

No Documented Dependency ≠ Demonstrated Independence

Strong Dependency ≠ High Materiality by Default

Mutual Support ≠ Independent Confirmation

Dependency Breach ≠ Automatic Assumption Failure

Current Dependency Structure ≠ Historical Dependency Structure

Enterprise Application

ADMM enables organizations to see whether predictive assumptions are truly independent or merely different expressions of the same underlying dependency.

This becomes critical in:

AI forecasting + supply-chain prediction + financial modeling + industrial systems + strategic planning + insurance + healthcare + infrastructure + public policy + autonomous systems.

Without dependency mapping, a prediction may appear diversified while remaining structurally dependent upon:

one supplier; one data source; one infrastructure service; one regulatory state; one market condition; one human decision; one system assumption.

ADMM exposes these dependency structures before PAS evaluates sensitivity and materiality.

Use Case 1 — Five Assumptions Fail Together Because They All Depend on One Supplier

A manufacturer maintains five explicit predictive assumptions:

production capacity remains available; component supply remains stable; delivery schedules remain achievable; inventory remains above minimum threshold; customer orders can be fulfilled.

The assumptions appear separate.

Dependency mapping reveals that all five materially depend upon one overseas supplier providing a critical component.

The supplier represents a single upstream dependency supporting much of the prediction.

The critical governance question is:

Can five documented assumptions legitimately be treated as five independent predictive conditions when all of them materially depend upon one underlying supplier relationship?

ADMM maps the shared upstream dependency and the downstream assumption chain.

Later PAS modules can then determine how sensitive and material this concentration is.

Multiple Assumptions ≠ Independent Assumptions

Use Case 2 — An AI Forecast Contains a Circular Assumption That Reinforces Itself

An AI system predicts continued growth in an investment market.

Its reasoning depends partly upon:

investor confidence remains strong because investment continues.

At the same time, another assumption states:

investment continues because investor confidence remains strong.

Each assumption appears to support the other.

Without dependency mapping, the system may present the circular relationship as two reinforcing pieces of predictive support.

The critical governance question is:

Can mutually dependent assumptions legitimately be interpreted as independent support when each derives its plausibility from the other?

ADMM identifies the circular dependency.

The prediction can then preserve the relationship as a structural assumption condition rather than treating it as independent confirmation.

Mutual Support $\neq$ Independent Confirmation

Canonical Boundary

ADMM governs the identification, classification, direction, structure, concentration, state, uncertainty, historical preservation, and propagation pathways of material dependencies among Predictive Assumptions and between assumptions and relevant inputs, methodologies, conditions, systems, decisions, resources, and other predictive elements. It does not govern assumption identification itself, Assumption Sensitivity, Assumption Materiality, Assumption Change Governance, evidence validity, Operational Reality, Prediction Formation, probability representation, or prediction-level revision.

ADMM establishes:

what each assumption depends upon and what may depend upon it.

It does not determine:

how strongly the prediction will change when that dependency changes.

Canonical Closing Statement

Assumptions do not become independent merely because they are written on separate lines.

ADMM reveals the dependency structure beneath the Prediction Assumption Set so that a single hidden dependency cannot masquerade as multiple independent predictive conditions.

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